Interactively Visualizing Multivariate Market Segmentation Using the R Package Lionfish: Response to Review

We sincerely thank the editor for your valuable suggestions, which have greatly improved the quality of both the article and the lionfish software.

General remark

The following summaries all of our changes and explanations to the comments provided. *The editor's comments are shown in italic.* When a comment resulted in specific changes in the text, we have added the revised text to the response to that comment.

Editor

General Comments

The abstract and introduction should be improved to

(1) focus more on clustering results obtained using k-means clustering where a partition of the space is obtained which can be obtained using linear combinations of features,

Response: The tone of the text was adjusted to focus more on results obtained with k-means.

(2) add more references to other visualization techniques for clustering solutions, data structure analysis in clustering and the importance of visualizations for market segmentation, in particular those by Fritz,

Response: The suggested references are now included.

(3) add an overview on the structure of the manuscript at the end of the introduction where you outline the different sections and what they provide.

Response: A brief overview is now given at the end of the introduction.

Specific Comments

- *Regarding (1), please note that one implicitly already focuses on such clustering results and some claims made would not hold for other clustering approaches used, e.g., auto-encoders, model-based approaches, etc. Later on only k-means clustering results are considered and this represents the default clustering method in many applications anyway. So it would ease the presentation to focus on this clustering method first and later on discuss extensions to other clustering approaches.*

  Response: This is made clearer in the abstract and introduction now.

  Added text (Abstract): The focus is on the widely used k-means clustering algorithm, but the tools will support other algorithms, also.

  Added text (Intro): In this article, we will (i) describe the lionfish software and (ii) demonstrate its utility through three example analyses in the domain of market segmentation using k-means clustering algorithms.

- *Regarding (2), please have a look at:*

  Nazila Babakhani, Friedrich Leisch, and Sara Dolnicar. A good graph is worth a thousand numbers. Annals of Tourism Research, 76:338-342, 2019. https://dx.doi.org/10.1016/j.c

  Friedrich Leisch. Neighborhood graphs, stripes and shadow plots for cluster visualization. Statistics and Computing, 20(4):457-469, 2010. https://dx.doi.org/10.1007/s11222-009-9137-8

  Friedrich Leisch. Visualizing cluster analysis and finite mixture models. In Chun-houh Chen, Wolfgang Härdle, and Antony Unwin, editors, Handbook of Data Visualization, Springer Handbooks of Computational Statistics, pages 561-588. Springer Verlag, 2008.

  Sara Dolnicar and Friedrich Leisch. Evaluation of structure and reproducibility of cluster solutions using the bootstrap. Marketing Letters, 21:83-101, 2010. https://dx.doi.org/10.10
  009-9083-4

Christian Hennig. Asymmetric Linear Dimension Reduction for Classification. Journal of Computational and Graphical Statistics, 13(4):930-945, 2004. https://doi.org/10.1198/

Response: References added.

Added text Introduction: ...and the advantage of visualization for the interpreation of the resulting clusters was demonstrated in Babakhani, Leisch, and Dolnicar (2019).

In a realistic, higher-dimensional setting, dedicated approaches for the visualization of cluster analysis Leisch (2008) can provide more detailed insights. For example, neighbourhood graphs Leisch (2010) can be used in combination with linear projections, to include information about cluster separation. This is in particular interesting when looking for projections that separate one of the clusters from the rest, as possible when using the methods suggested in Hennig (2004). The resulting view will often show other clusters as overlapping, even when they are clearly separated in the full space.

Added text Reproducibility: Note that here the focus is on the reproducibility of the interactive data analysis. Reproducibility of the cluster solution can be evaluated using a bootstrap approach Dolnicar and Leisch (2010).

- *Regarding (3), it might be good to emphasize the general applicability of the interactive tools provided and that the segmentation analysis is only one possible area of application which is, however, of focus in this contribution with the different illustrative applications. Note that Section 2 seems to be rather generic and not provide much information on use for inspecting clustering solutions, e.g., Section 2.1 does not mention any clustering results as input. So guiding readers on the different sections and what they present could help understanding.*

Response: The first paragraph has been revised to be more specific on the startup, and about the general applicability.

New/changed text: The lionfish GUI can be launched using the function `interactive_tour()`. At minimum, this function requires the data and a list of plot objects to display (typically a 1D or 2D tour, a barchart, scatterplot, etc.). Optionally, the user can define an initial clustering of the data as cluster memberships matching the rows of the data, the layout of the plots, the number of available subsets, the size of the GUI, initially selected features for display, a scaling factor for the projected data and a few minor plotting instructions such as used color schemes. If no initial clustering is provided all data will be in one subset, and partitioning can be performed later via manual selection. The `n_subsets` argument fixes the number of available partitions of the GUI upon startup. If `n_subsets` is larger than the number of initially provided partitions, then the extra partitions will be empty, but can be filled via manual selection. This can be useful when a user wants to shift some data of special interest (e.g. outliers) into an extra set or redefine parts of a clustering solution as shown in the case study in Section 4.2. All features in the data are expected to be numeric, so categorical features should be converted prior to using the GUI. If they are not, then the conversion will be automated in the data pre-processing, for example, a variable called "fruit" with categories "banana", "orange", "peach" will be recoded as 1, 2, 3, respectively. The plotting instructions for each display to be shown have to be provided in form of a named list containing a `type` and an `obj`. The `type` element specifies the type of display to generate, such as `scatter` for a scatterplot or `2d_tour` for a 2D tour. The `obj` element further defines the properties of the chosen display. For example, to create a 2D tour, the user must provide a `tour_history` object, which can be generated using the tourr package. For a scatterplot, the user needs to provide a character vector specifying the names of the features to be displayed. In this article, the focus is on the use of lionfish for exploring k-means clustering results in context of market segmentation, but the package website contains examples of usage for other applications, animated examples showing the interactivity, and details on function arguments.

Added text: Section 4 now starts with "In this section, the Austrian Vacation Activities (Dolnicar and Leisch 2003), Australian Vacation Activities (Cliff 2009) and Tourist Risk Taking (Dolnicar 2017) datasets are analyzed with lionfish to illustrate the software's capabilities. It has to be underlined that these analyses only serve as illustrative case studies and that the application of lionfish is not limited to market segmentation."

Further Comments

- *Refer to the package being on CRAN including the URL (when first mentioning and in the Section "Resources and supplementary materials").*

  Response: Done.

- *Please try to consistently refer to clusters / segments and avoid also using the term groups.*

  Response: All instances of "groups" has been changed to "subsets". The term cluster is used only in relation to clustering algorithms. The term "segments" is replaced with "partitions" to indicate the cutting of the data into subsets, when there are no (substantial) gaps between points. "Groupings" is used twice when it seems more appropriate than clustering or partitioning.

- *Please explain what "bit blit control" is as a reader might not be familiar with this and decide to write "bit blit" or "bit-blit".*

  Response: The opening paragraphs of Section 2 now includes a pointer to the explanation of "bit blit" in Section 2.4, and "bit blit" is used throughout.

  Revised text (opening of Section 2): Through matplotlib, Python provides interactivity on the plots, allowing elements to be directly selected or changed, and low-level bit blit control (for details see section 2.4), enabling efficient updates and rendering of plots.

  Revised text (Section 2.4): With bit blit, the static elements of the display, such as the outer frames of the plots, are stored as a background image. When a plot is manipulated, only the affected plot is updated, and within that, only the interactive elements, such as the datapoints and projection axes in a 2D tour, are rendered on top of the background image.

- *It is unclear what is meant with "to numeric values that reflect the categories". Please be more specific.*

  Response: The paragraph was changed (see above) which gives more context, and additional explanation was added.

- *On page 9, the extension to other feature types is unclear, e.g., how are feature means determined for ordered categories.*

  Response: Maybe this is clearer now given the improved description of the transformation of categorical variables to numeric values?

- *Regarding the importance of reproducibility one might also refer to other contributions in this SI:*

  Roger D. Peng. Tooling for Reproducible Research: Considerations for the Past and Future of Data Analysis. Austrian Journal of Statistics. doi:10.17713/ajs.v54i3.2052

  Robert Gentleman, Anthony Rossini and Vincent Carey: Contributions of Fritz Leisch to Vignettes and Reproducible Research. Austrian Journal of Statistics. doi:10.17713/ajs.v54i3.2057

  Response: Done

- *Note that Likert scales usually are referred to as multi-item scales where each item is evaluated on an ordinal scale ranging from disagree to agree and where an aggregate score across items is determined. Your data example hence thus not seem to include Likert data. You have ordinal/ordered categorical variables.*

  Response: "Likert" was changed to "ordinal" throughout.

- *Not sure about "country's capital cities" as Austria clearly only has a single capital city or did you mean not country but federal state?*

  Response: In Dolincar, 2003 is written: "Please note that tourists staying in the capital cities of Austria are not included in this sample, as a different questionnaire was used for this group." Because it is unclear we have removed this part of the sentence.

- *One might align the clusters in Figure 7 using classAgreement from package e1071 authored by Fritz Leisch.*

  Response: classAgreement is now used to compare the similarity between the cluster solutions.

  Revised text: We can now repeat the k-means clustering with stepcclust() on the reduced dataset. To evaluate the similarity between the two cluster solutions, we can use the e1071 package's classAgreement() function (Meyer, Dimitriadou, Hornik, Weingessel, and Leisch 2024). This function, among other metrics, calculates κ, the percentage of data points that are in the same cluster adjusted for chance. The resulting κ value of 0.56 suggests that while there is some overlap between the cluster solutions, they differ substantially. Silhouette plots of both cluster solutions can be seen in Figure 7. By comparing both silhouette plots, we can see that the cluster solution with the reduced dataset results in a clustering of higher quality. Thus, we will continue with the analysis on the reduced dataset with the corresponding cluster solution. We can also see in Figure 7B that cluster 3 is of comparatively low quality.

- *The purpose of the paragraph "It is important to note that the silhouette scores ..." is rather unclear as well as its fit at this point in the manuscript.*

  Response: The purpose of this paragraph is to further underline, based on an actual example, that clustering solutions are not always of high quality and useful when dealing with diffuse datasets, and that introducing expert knowledge can improve the analysis.

- *The comment about removing duplicates in the risk data set is unclear. Given that the data consists of answers on 6 variables with 5 categories, it is unclear why respondents who give the exact same answers are to be removed. It would seem that these duplicates still represent genuine observations.*

  Response: The analysis has been performed again without deletion of the duplicates.

Stylistic issues

- *Please consistently use either British or American English.*

  Response: The manuscript is now in American English exclusively.

- *Please use consistent names for the data set (including consistent lower/upper case). Also the dash in "Risk-taking" is unclear.*

  Response: Done.

- *Omit numbers from Sections 7 and 8.*

  Response: Done.

- *Use pkg mark-up for packages as defined in ajs.cls.*

  Response: Done.

- *Please use "datapoints" or "data points".*

  Response: Only "datapoints" is used now.

- *Check when abbreviations (e.g., GUI) are introduced the first time and then consistently use them.*

  Response: Done.

- *Use suitable markup for classes and indicate character strings correctly.*

  Response: The manual projection manipulation classes and selector classes are now explicitly mentioned and within texttt.

  Revised text: Plot-specific data is organized in dictionaries (the Python equivalent of named lists in R), including the display type, construction instructions, tour projections (only in case of tour displays), color schemes for the displayed data, and, where applicable, the selector classes (`BarSelect` and `LassoSelect`) and manual projection manipulation classes (`DraggableAnnotation1d` and `DraggableAnnotation2d`).

- *Use consistent markup for file extensions.*

  Response: texttt is now used for all file extensions.

- *Figure missing in "as seen in 13".*

  Response: Added "Figure".

- *Use a consistent labeling for "Guided tour - LDA - regroup".*

  Response: Instances of "Guided tour - LDA - Regroup" were changed to "Guided tour - LDA - regroup"